



FACULTY OF COMPUTING
SEMESTER 1 2025/2026

SECJ 3553 - ARTIFICIAL INTELLIGENCE
SECTION 03

AI Project - Assignment 1

THEME:
SMART CITY - ECO-ZONE AIR CONDITIONER

LECTURER: DR. NORSHAM BINTI IDRIS

STUDENT NAME	MATRIC NO
CHAU YING JIA	A23CS0213
NURUL ASYIKIN BINTI KHAIRUL ANUAR	A23CS0162
NEO LI XIN	A23CS0253
AFIF SHAQIR IRFAN BIN ARQAM	A23CS0204

Introduction

Our project aim is to develop an intelligent AI system to minimize energy waste while enhancing student comfort in the UTM library. To achieve the goals outlined by our Eco-Zone Air Conditioner system, our AI will need a 'brain' capable of making intelligent decisions. This brain comprises its Knowledge Representation, which consists of a set of formal, logical rules.

These rules will convert our defined user needs into specific actions that the air conditioning system should undertake. Essentially, our AI will rely on these rules to determine the optimal temperature for each zone by analysing data gathered from the sensors.

Predicates

Prior to codifying our knowledge representation rules, it's necessary to define the underlying facts or predicates our AI program would base its reasoning about the library's environment. These predicates form the vocabulary through which our AI perceives and interacts with the world. The variable z signifies a given zone of the library.

1. IsEmpty(z)
Definition: True if the motion sensor in zone z detects no people.
2. IsOccupancyHigh(z)
Definition: True if occupancy in zone z is 10 or more people.
3. IsRaining()
Definition: True if external weather sensor detects rain or outdoor temperature is below 24°C.
4. IsNight()
Definition: True if current time is after 10 PM.
5. OutdoorTemp(temp)
Definition: True if external outdoor temperature equals temp.
6. PredictedOccupied(z, t)
Definition: True if machine learning model predicts zone z will be occupied at future time t .
7. SetTemp($z, temp$)
Definition: The AC system is instructed by the AI to set zone z to temp.

Knowledge Representation for AI Solution

Rule 1: Air Conditioner Eco-Mode Rule (Supports Need 2)

This rule specifically targets the need to reduce the energy costs outlined in our proposal. It has the potential to optimize power usage by setting the target temperature to 24°C in the empty rooms with the "Eco Mode".

First-Order Logic (FOL):

$$\forall z \text{ IsEmpty}(z) \rightarrow \text{SetTemp}(z, 25)$$

Explanation:

For every zone z in the library, if that specific zone z is empty, then set the temperature for that zone z to 25°C.

Rule 2: Weather-Adaptive Temperature Rule (Supports Need 3)

This rule supports the need to respond to outdoor weather conditions. It ensures that when the external temperature drops, the system automatically increases the indoor temperature slightly to maintain user comfort and prevent overcooling.

First-Order Logic (FOL):

$$\forall z \forall \text{temp} (\text{IsRaining} \vee \text{OutdoorTemp}(\text{temp}) < 24) \rightarrow \text{SetTemp}(z, 24)$$

Explanation:

For every zone z in the library, if it is raining or if the outdoor temperature is below 24°C, then the system should increase the indoor temperature to 24°C for that zone to maintain comfort while conserving energy.

Rule 3: Predictive Pre-Cooling Rule (Supports Need 1)

This rule addresses the need for a comfortable study environment. It utilizes historical occupancy data to pre-cool zones that are predicted to have higher user presence, ensuring an ideal temperature before users arrive.

First-Order Logic (FOL):

$$\forall z \forall t (\text{PredictedOccupied}(z, t+1)) \rightarrow \text{SetTemp}(z, 22)$$

Explanation:

For every zone z in the library, if the system predicted that zone to be occupied at the next time stamp $(t+1)$, then set the temperature to 22°C to pre-cool the environment for incoming users.

Rule 4: Night-Time Energy Saving Rule (Supports Need 2)

This rule ensures the system reduces cooling during late-night hours when student presence is minimal, helping cut unnecessary energy usage.

First-Order Logic (FOL):
$$\forall z (\text{IsNight}() \wedge \text{IsEmpty}(z)) \rightarrow \text{SetTemp}(z, 25)$$
Explanation:

For every zone z , if it is night time *and* that zone is empty, the system increases the temperature to 25°C to minimize energy wastage since fewer students remain in the library during late hours.

Rule 5: High Occupancy Comfort Rule (Supports Need 1)

This rule handles zones that are crowded. High occupancy generates more body heat, so the system must cool more aggressively to maintain comfort.

First-Order Logic (FOL):
$$\forall z \text{IsOccupancyHigh}(z) \rightarrow \text{SetTemp}(z, 20)$$
Explanation:

For every zone z , if the occupancy level is high (10 people or more), the system will set the temperature to 20°C to counteract the extra heat generated by the crowd and ensure a comfortable environment.

Conclusion

In summary, the defined predicates and FOL rules allow the AI to respond intelligently to occupancy patterns, weather conditions, and time of day while balancing both comfort of users and energy efficiency. With this implementation of knowledge representation, the system is prepared for further development and integration by formulating the problems using state space search.